HACKATHON PROJECT

Can RSI improve an LLM's understanding of physics?

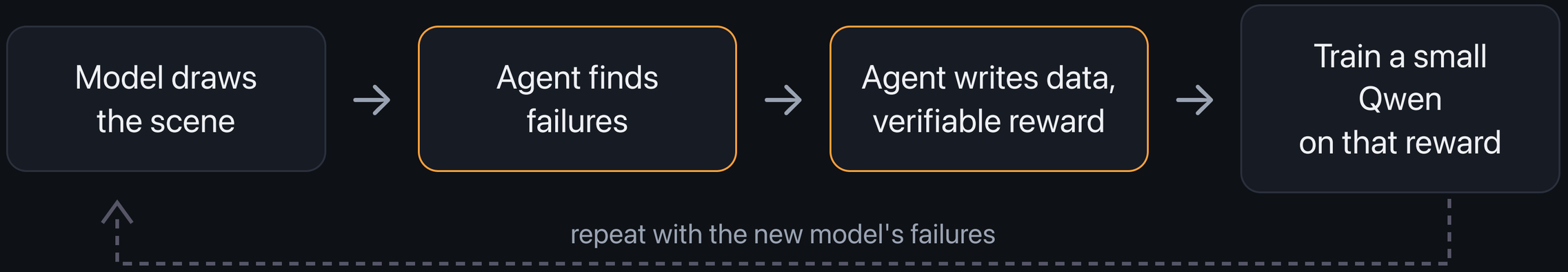
THE PROBLEM

AI-generated scenes look real, but their motion often breaks the physics.

Newton's laws. Energy conservation. Momentum.

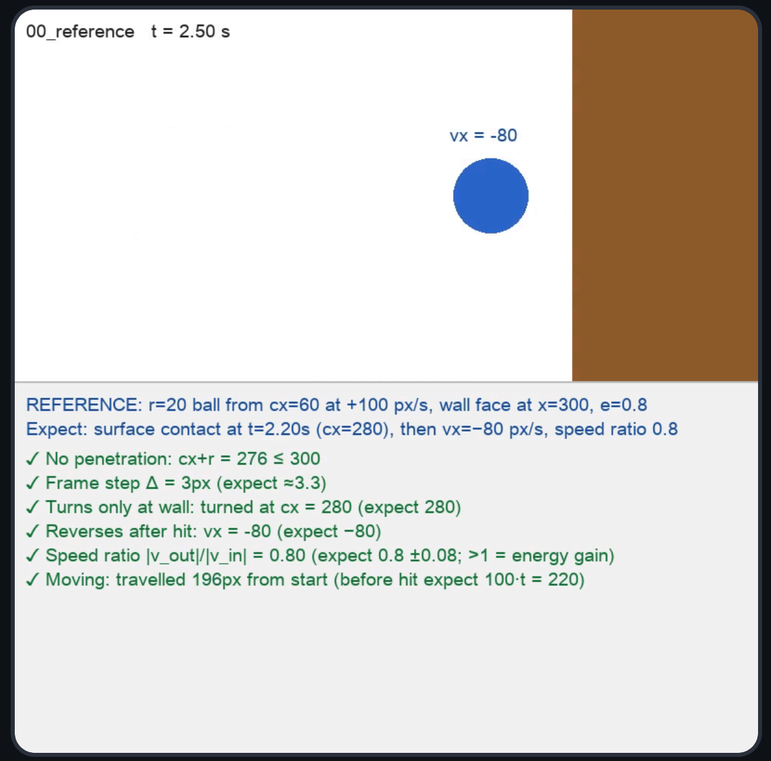
OUR IDEA

Put an agent in the improvement loop.



EXAMPLES

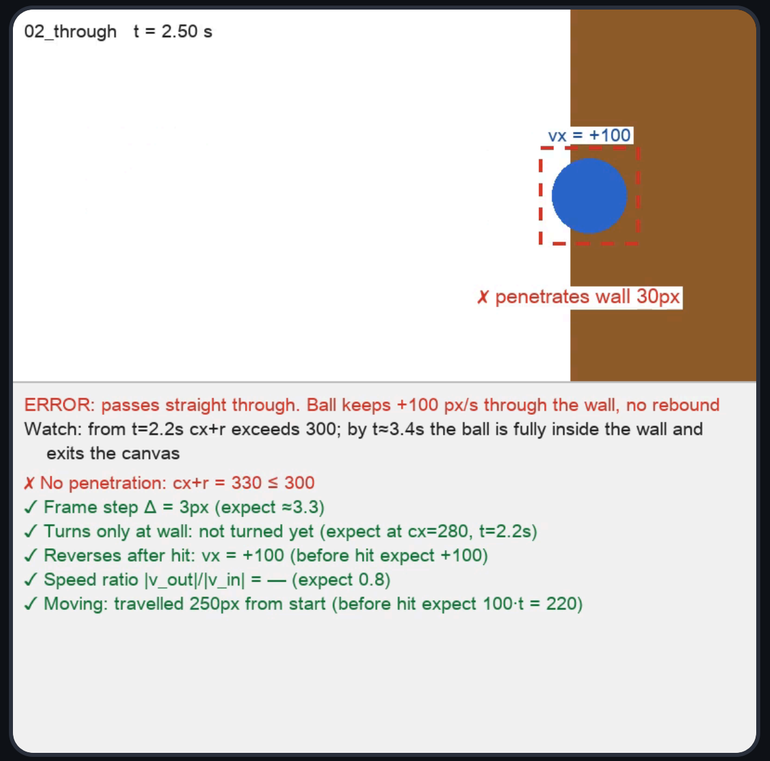
A correct bounce, and a failure we catch.



REFERENCE: r=20 ball from cx=60 at +100 px/s, wall face at x=300, e=0.8
Expect: surface contact at t=2.20s (cx=280), then vx=−80 px/s, speed ratio 0.8

✓ No penetration: cx+r = 276 ≤ 300
✓ Frame step Δ = 3px (expect ≈3.3)
✓ Turns only at wall: turned at cx = 280 (expect 280)
✓ Reverses after hit: vx = -80 (expect −80)
✓ Speed ratio |v_out|/|v_in| = 0.80 (expect 0.8 ±0.08; >1 = energy gain)
✓ Moving: travelled 196px from start (before hit expect 100·t = 220)

Correct



ERROR: passes straight through. Ball keeps +100 px/s through the wall, no rebound
Watch: from t=2.2s cx+r exceeds 300; by t≈3.4s the ball is fully inside the wall and exits the canvas

✗ No penetration: cx+r = 330 ≤ 300
✓ Frame step Δ = 3px (expect ≈3.3)
✓ Turns only at wall: not turned yet (expect at cx=280, t=2.2s)
✓ Reverses after hit: vx = +100 (before hit expect +100)
✓ Speed ratio |v_out|/|v_in| = — (expect 0.8)
✓ Moving: travelled 250px from start (before hit expect 100·t = 220)

Failure → becomes a reward signal

RESULTS

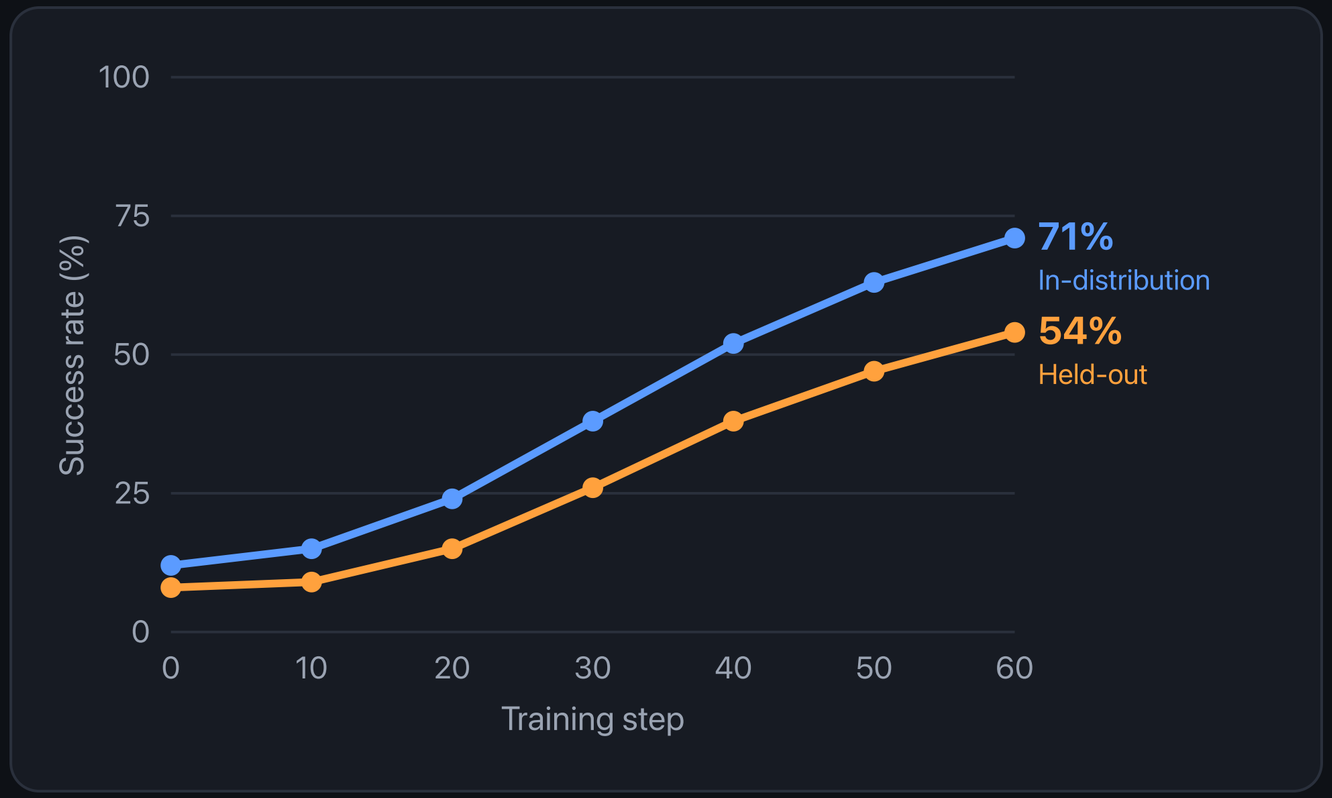
Success rate on cases it never saw.

8% → 54%

Held-out scenes, before → after training

12% → 71%

In-distribution scenes



TAKEAWAY

Recursive self-improvement, for physics.